

ATROPOS OBLIGATION DAG

Source Document 7 → Hybrid RESEARCH+IMPL + Track/WIRE

Tier-1 Priority: Complete Extirpation of AppSourceTemplate.kt & LanguageScaffold.kt

Maximum atomization · Zero artificial templates · Every atom independently verifiable · 2026-10-06

1. Extraction Rules (Locked)

Maximum atomization. Blueprint phases do not consume concrete atoms. Every feature, function, utility, invariant, gate, driver, and mathematical claim is decomposed until it cannot be divided further.

Hybrid format: Residual Obligation DAG v3 RESEARCH (R:) + IMPL (I:) density combined with HOE Track / dependsOn / WIRE discipline.

Research prompts force external discovery of superior implementations; they never restate Source Doc 7 sketches.

Impl notes name foreseeable failure modes and the cheapest correct path that preserves existing ATROPOS substrate (IsolatedWorktreeService, VerifiedCompletionGate, BoundedProcessRunner, TerritoryGrant, ArchitectureComplianceChecker, SpecGraphCompiler, DLOI).

Status tags: RESIDUAL | AMENDMENT-CANDIDATE | NEW-SUBSTRATE | 2050-NEURO-SYMBOLIC | SUPERSEDED-CAPABILITY.

WIRE may name classes newly introduced by Source Doc 7. Final section of this document is a mandatory Caller-Completeness Gate that must pass before any promotion.

Tier discipline: Tier-1 (this document) must reach verified completion before any subsequent tier is scheduled. Tier-1 = total purge of static templates + live Interface-First Wave factory that produces real applications.

2. Tier Structure

Tier-1 (this DAG): Extirpation of AppSourceTemplate.kt / LanguageScaffold.kt + full Wave 0–4 factory substrate + native toolchains + 2050 neuro-symbolic gates. Acceptance = real applications compile and pass VerifiedCompletionGate with zero template bytes.

Tier-2 (future): Long-horizon storage constitution, swarm hierarchy, thermal/battery pacing, exact-hash restart continuity, quota ledger.

Tier-3 (future): Full CLI/Unix sovereignty, PTY process hosting, shell completion engine, OSC-52 clipboard, anti-fake-green oracles at production scale.

Tier-4 (future): Cloud delivery drivers (Vercel/Netlify/Supabase) + packaging to bit-verified ELF/APK/Web.

3. Tier-1 Atoms — Template Extirpation & Core Factory Substrate

3.1 Deprecation & Routing Severance

SD7-T1-001+extirpate-AppSourceTemplate [NEW-SUBSTRATE]

Track: S (Shared Core)

dependsOn: []

R: Research the complete call graph of AppSourceTemplate.kt and every consumer that imports mainSource(), testSource(), genericMain() or any other string-literal emitter. Identify the minimal set of deletion points that leave zero residual template paths.

I: Hard-delete the file from the critical path. Remove every import and call site. Replace with a single compile-time assertion that fails if any template string literal matching the old signatures is present in the factory package. Foreseeable error: residual string constants surviving in test fixtures or comments; scan with ArchitectureComplianceChecker.

WIRE: AppFactoryRouter.kt (mutate plan() and runLocal()), ArchitectureComplianceChecker (new ban rule), DagExecutionService.kt

SD7-T1-002+extirpate-LanguageScaffold [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-001]

R: Research every LanguageScaffold entry point and language-specific string templates. Locate any dynamic dispatch that still routes to scaffold methods.

I: Hard-delete. All language-specific scaffolding is replaced by Wave-0 ContractLatticeSynthesizer output. Foreseeable error: language-kind switch statements that still call scaffold; convert them to throw UnsupportedOperationException until Wave-0 contracts exist.

WIRE: AppProjectGenerator.kt, AppFactoryRouter.kt

SD7-T1-003+sever-AppFactoryRouter [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-001, SD7-T1-002]

R: Research clean router patterns that completely detach intent parsing from code emission. Examine how modern agent factories (Aider, Codex, OpenCode) keep planning and synthesis orthogonal.

I: Mutate AppFactoryRouter so plan() and runLocal() emit only a typed WaveExecutionRequest and hand control to AutonomousWaveOrchestrator. No template path may remain reachable. Foreseeable error: fallback branch that still calls generateApp() with templates; delete the fallback.

WIRE: AutonomousWaveOrchestrator (new), AppFactoryRouter.kt

SD7-T1-004+mutate-DagExecutionService [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-003]

R: Research batched topological wave processors that respect territory isolation and worktree lifecycle. Prefer designs that never hold the entire atom set in a single context window.

I: Replace synchronous execution stubs with wave-batch processing. Every PROVIDER_CALL node must execute inside an IsolatedWorktreeService worktree. Foreseeable error: shared mutable state across partitions; enforce territory disjointness before dispatch.

WIRE: DagExecutionService.kt, IsolatedWorktreeService, PartitionClusterManager

3.2 Wave-0 Substrate & Toolchains

SD7-T1-010+NativeToolchainRegistry [NEW-SUBSTRATE]

Track: S

dependsOn: []

R: Research deterministic toolchain discovery patterns used by Bazel, Pants, and Nix that locate compilers without hard-coded absolute paths and survive Termux / Linux / macOS differences.

I: Implement NativeToolchainRegistry that probes PATH and known locations, returns typed LanguageToolchainDriver instances, and never swallows non-zero exit codes. Foreseeable error: silent fallback to mock drivers; ban any driver that does not invoke a real binary.

WIRE: ToolchainExecutionDriver, BoundedProcessRunner, VerifiedCompletionGate

SD7-T1-011+LanguageToolchainDriver-interface [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-010]

R: Research minimal interface contracts that cover compile, test, lint, package for heterogeneous languages while remaining extensible to new drivers.

I: Define LanguageToolchainDriver with compile(worktree), test(worktree), lint(worktree), package(worktree). All methods return structured Diagnostic maps (file, line, column, code, message). Foreseeable error: drivers that return success on empty worktrees; require at least one source file.

WIRE: NativeToolchainRegistry, all concrete drivers

SD7-T1-012+RustCargoDriver [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T1-011]

R: Research cargo check / cargo test / cargo build --release exit-code and diagnostic capture under bounded process runners.

I: Implement via BoundedProcessRunner. Capture stdout/stderr, map to Diagnostic. Non-zero exit = terminal failure. Foreseeable error: interactive prompts hanging the process; always supply --color=never and non-interactive flags.

WIRE: NativeToolchainRegistry, ToolchainExecutionDriver

SD7-T1-013+TypescriptNodeDriver [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T1-011]

R: Research pnpm/npm + tsc --noEmit + vite build diagnostic patterns and lockfile isolation.

I: Run inside isolated worktree. Use --frozen-lockfile when possible. Non-zero tsc or build exit fails the gate. Foreseeable error: node_modules leakage across worktrees; always install into the worktree only.

WIRE: NativeToolchainRegistry

SD7-T1-014+GoToolchainDriver [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T1-011]

R: Research go build / go vet / go test -race under resource limits.

I: BoundedProcessRunner with explicit GOMAXPROCS and timeout. Race detector enabled for test. Foreseeable error: module cache pollution; set GOPATH/GOCACHE inside worktree.

WIRE: NativeToolchainRegistry

SD7-T1-015+PythonFastApiDriver [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T1-011]

R: Research mypy --strict + ruff + pytest isolation patterns for FastAPI projects.

I: Create ephemeral venv inside worktree if needed. mypy --strict must pass before pytest. Foreseeable error: system site-packages contamination; always --isolated.

WIRE: NativeToolchainRegistry

SD7-T1-016+KotlinGradleDriver [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T1-011]

R: Research ./gradlew compileKotlin test assemble under daemon control and offline mode.

I: Prefer --offline when caches are present. Capture Gradle diagnostic XML if available. Non-zero exit fails. Foreseeable error: Gradle daemon leaking across worktrees; use --no-daemon or unique project cache.

WIRE: NativeToolchainRegistry, GovernedCompileGate

SD7-T1-017+AndroidSdkDriver [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T1-011, SD7-T1-016]

R: Research AAPT2 + D8/R8 + gradlew assembleDebug production of unsigned APK with bit-level verification.

I: Produce real APK artifact. Verify APK exists and has non-zero size. Foreseeable error: missing ANDROID_HOME; fail closed with clear diagnostic rather than mock APK.

WIRE: NativeToolchainRegistry, CloudDeploymentAdapter (later tier)

SD7-T1-018+PartitionClusterManager [NEW-SUBSTRATE]

Track: S

dependsOn: []

R: Research Kahn topological sort + greedy domain-path proximity clustering under token budgets. Examine how modern build systems (Bazel, Pants) partition for parallel execution while preserving acyclicity.

I: Implement Ψ operator. Constraints: quotient graph remains DAG, sum of tokens + interface context $\leq B_{\text{token}}$ (default 12 000), territories pairwise disjoint. Foreseeable error: cycles introduced by greedy merge; re-run topological check after every cluster formation.

WIRE: InternalExecutionDagSynthesizer, AutonomousWaveOrchestrator, TerritoryGrant

3.3 Wave-0 Contract Lattice

SD7-T1-020+ContractLatticeSynthesizer [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-010, SD7-T1-018]

R: Research pure interface / trait / schema synthesis that emits zero executable statements. Examine OpenAPI 3.1, Protobuf, SQL DDL, TypeScript .d.ts, Rust traits, Kotlin interfaces for minimal complete contracts.

I: Wave-0 output contains only type signatures, schemas, and protocol definitions. Compilation gate is typecheck/lint only. Repair until clean. Foreseeable error: accidental inclusion of function bodies; AST walk rejects any executable node.

WIRE: AutonomousWaveOrchestrator, DecoupledWorkerSynthesizer, ArchitectureComplianceChecker

SD7-T1-021+clean-room-prompt-contract [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-020]

R: Research prompt formats that mathematically enforce $I(f;fk | I) = 0$ by construction. Examine context isolation techniques used in secure multi-party computation and clean-room design reviews.

I: AgentPromptContract must inject the Wave Contract format. Worker receives only atom payload + Wave-0 interfaces. Sibling implementation text is never present. Foreseeable error: accidental inclusion of previous worker diffs; scrub context before every dispatch.

WIRE: AgentPromptContract.kt, DecoupledWorkerSynthesizer

3.4 Wave-1 Behavioral Verification Suites

SD7-T1-030+Wave1-test-suite-generator [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-020, SD7-T1-021]

R: Research property-based and interface-mock test generation that must fail against empty stubs and compile against Wave-0 contracts only.

I: Generate isolated unit + property tests. Gate: tests compile against contracts and fail on empty implementations. Foreseeable error: tests that pass vacuously; require at least one failing assertion on a stub.

WIRE: ContractLatticeSynthesizer, VerifiedCompletionGate

3.5 Wave-2 Parallel Partition Implementation

SD7-T1-040+DecoupledWorkerSynthesizer [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-018, SD7-T1-021, SD7-T1-030]

R: Research isolated worktree synthesis loops with bounded repair (max 3) that feed compiler diagnostics back without expanding context beyond the partition footprint.

I: Each partition runs in its own IsolatedWorktreeService worktree. Worker sees only its atoms + Wave-0 contracts. On failure, repairNode feeds stderr back. Merge only after clean verification. Foreseeable error: territory collision; fail closed and revert worktree.

WIRE: IsolatedWorktreeService, GovernedCompileGate, ProviderCascadeRouter, PartitionClusterManager

SD7-T1-041+CompilerDiagnosticRepairLoop [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-040]

R: Research diagnostic-to-prompt feedback loops that preserve the original atom and interface set while only mutating the failing file.

I: Max 3 attempts. Each attempt receives the exact compiler diagnostic map. No expansion of context. Foreseeable error: infinite repair oscillation; after 3 failures create a directed RepairNode and halt the partition.

WIRE: DecoupledWorkerSynthesizer, GovernedCompileGate

3.6 Wave-3 DI & Composition

SD7-T1-050+DependencyInjectionAssembler [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-040]

R: Research pure DI registry synthesis (Koin, Dagger, NestJS modules, FastAPI Depends, Go wire) that never hard-codes sibling constructors.

I: Wire concrete implementations to Wave-0 interfaces only. Ban static singletons and service locators. Foreseeable error: direct new of sibling classes; ArchitectureComplianceChecker must reject.

WIRE: AutonomousWaveOrchestrator, ArchitectureComplianceChecker

3.7 Wave-4 Global Verification & Orchestration

SD7-T1-060+AutonomousWaveOrchestrator [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-003, SD7-T1-018, SD7-T1-020, SD7-T1-040, SD7-T1-050]

R: Research five-wave execution engines that treat each wave as a hard gate and never proceed on partial success.

I: Execute Wave 0 → 1 → 2 → 3 → 4. Any Halted report aborts the build with diagnostics. Success produces artifact path + optional deployment URL. Foreseeable error: soft failures that continue; every non-zero gate is terminal.

WIRE: AppFactoryRouter, PartitionClusterManager, ContractLatticeSynthesizer, DecoupledWorkerSynthesizer, DependencyInjectionAssembler, NativeToolchainRegistry

SD7-T1-061+VerifiedCompletionGate-toolchain-binding [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-010, SD7-T1-060]

R: Research binding of language-specific native exit codes directly into completion calculus so that fake success is impossible.

I: Mutate VerifiedCompletionGate to invoke the concrete LanguageToolchainDriver and treat any non-zero exit as terminal rejection. Foreseeable error: shell scripts that echo PASS; ban any verification that does not execute a real compiler binary.

WIRE: VerifiedCompletionGate.kt, NativeToolchainRegistry, BoundedProcessRunner

SD7-T1-062+mutate-ExecutionPolicyEngine [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-010]

R: Research policy engines that allow automated toolchain invocation inside verified sandboxes without human tokens while still enforcing territory and secret boundaries.

I: Permit cargo, go, pnpm, clang, zig, gradlew inside BoundedProcessRunner sandboxes. Secrets stripped from env. Foreseeable error: secret leakage into subprocess; always clean env.

WIRE: ExecutionPolicyEngine.kt, BoundedProcessRunner, TerritoryGrant

SD7-T1-063+mutate-InternalExecutionDagSynthesizer [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-018]

R: Research partition mapping that groups unbounded atoms into territorial clusters while emitting formal input/output interface ports and boundary hashes.

I: Implement Ψ . Emit partition metrics and handoff manifests. Foreseeable error: token budget overflow; reject partitions that exceed B_token.

WIRE: InternalExecutionDagSynthesizer.kt, PartitionClusterManager

SD7-T1-064+mutate-AgentPromptContract [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-021]

R: Research formal Wave Contract prompt formats that block implementation leakage by construction.

I: Inject the clean-room prompt. Block any sibling code. Foreseeable error: provider context windows that drop the interface lattice; place interfaces in high-order immutable slot.

WIRE: AgentPromptContract.kt, DecoupledWorkerSynthesizer

SD7-T1-065+mutate-SpecGraph-throughput [NEW-SUBSTRATE]

Track: B

dependsOn: []

R: Research streaming atom extraction that removes arbitrary token collection caps for large specifications.

I: Mutate apps/specgraph-foundry compiler `__init__.py` and `execution_dag.py` to increase throughput and emit partition metrics + boundary hashes. Foreseeable error: memory exhaustion on huge specs; stream rather than materialize.

WIRE: SpecGraphCompiler, InternalExecutionDagSynthesizer

4. 2050 Neuro-Symbolic & Scale-Gap Atoms (Tier-1 Inclusion)

4.1 Inverse Latent Specification Synthesis (ILSS)

SD7-2050-ILSS-01+adversarial-goal-expansion [2050-NEURO-SYMBOLIC]

Track: S

dependsOn: [SD7-T1-020]

R: Research adversarial counter-specification mining used in security threat modeling and formal methods (STRIDE, attack trees, TLA+ negative properties). Prefer techniques that produce machine-checkable $\neg S$ without requiring a second full LLM call per atom.

I: Before Wave-0, a critic produces Negative Specification $\neg S$ covering Byzantine faults, network partitions, malicious ingress, memory exhaustion, tail-latency SLAs. Every user goal G is paired with anti-goal $\neg G$ that must provably never occur. Foreseeable error: open-ended questions; instead compute Pareto frontier and request single-bit human selection.

WIRE: ContractLatticeSynthesizer, DloiService, FactoryClarificationRequest

SD7-2050-ILSS-02+DLOI-domain-contract-extrapolation [2050-NEURO-SYMBOLIC]

Track: S

dependsOn: [SD7-2050-ILSS-01]

R: Research automatic grafting of canonical non-functional contracts from knowledge corpora (OWASP, ACID isolation proofs, connection-pool bounds, GC pressure invariants).

I: Query DLOI address map (006 Security, 410 Data Plane, 620 Runtime) and inject mandatory Wave-0 atoms regardless of user prompt brevity. Foreseeable error: over-injection that violates territory; keep injected contracts inside the same partition territory.

WIRE: DloiService, ContractLatticeSynthesizer, SourceAuthorityIndexer

SD7-2050-ILSS-03+epistemic-uncertainty-bounding [2050-NEURO-SYMBOLIC]

Track: S

dependsOn: [SD7-2050-ILSS-01]

R: Research entropy measures $H(\text{Intent}|\text{Prompt}, \text{DLOI})$ and Pareto-optimal decision matrices for under-determined goals.

I: If entropy $> \epsilon$, compute Pareto frontier (latency vs memory, consistency vs availability, throughput vs complexity) and await single-bit human selection before DAG freeze. Foreseeable error: silent default choices; always surface the matrix.

WIRE: FactoryConfidence, FactoryClarificationRequest, AutonomousWaveOrchestrator

4.2 Formal Concurrency Model Checking

SD7-2050-FMC-01+TLA-Alloy-state-machine [2050-NEURO-SYMBOLIC]

Track: S

dependsOn: [SD7-T1-020]

R: Research extraction of TLA+ / Alloy specifications from interface contracts and exhaustive checking with TLC or Z3 before any implementation code is written.

I: Every stateful or concurrent atom must emit formal state variables, transition relations, safety and liveness properties. Gate: exhaustive exploration proves zero deadlocks and zero race conditions. Foreseeable error: state-space explosion; bound the model and fail closed if incomplete.

WIRE: ContractLatticeSynthesizer, VerifiedCompletionGate

SD7-2050-FMC-02+chaos-digital-twin [2050-NEURO-SYMBOLIC]

Track: B

dependsOn: [SD7-T1-060, SD7-T1-061]

R: Research ephemeral containerized chaos runners that inject kernel preemption, socket latency, clock skew, and 95 % memory ceilings while measuring residual memory and error rates.

I: Wave-4 instantiates Digital Twin. Acceptance requires survival of 10 000 chaos transactions with zero thread leaks and residual memory returning to baseline post-GC. Foreseeable error: non-deterministic chaos; seed all PRNGs and record the seed in the evidence bundle.

WIRE: AutonomousWaveOrchestrator, NativeToolchainRegistry, EvidenceCollector

SD7-2050-FMC-03+actor-territory-message-formalism [2050-NEURO-SYMBOLIC]

Track: S

dependsOn: [SD7-T1-018, SD7-T1-021]

R: Research affine / ownership type systems and disjoint mailbox formalisms that make shared mutable memory across territory boundaries a compile-time error.

I: All inter-actor communication typed as disjoint mailboxes. Violations caught at compile time. Foreseeable error: hidden shared state via statics; ArchitectureComplianceChecker bans static mutable state.

WIRE: ArchitectureComplianceChecker, TerritoryGrant, DecoupledWorkerSynthesizer

4.3 Neuro-Symbolic SMT Verification

SD7-2050-NSV-01+Hoare-logic-annotations [2050-NEURO-SYMBOLIC]

Track: S

dependsOn: [SD7-T1-040]

R: Research machine-checkable pre/post/invariant annotations across Rust (Kani/Creusot), Kotlin, TypeScript, Python (icontract).

I: Every synthesized function must emit requires / ensures / invariant. Functions lacking provable contracts are rejected as unverified. Foreseeable error: annotations that are comments only; require toolchain that discharges them.

WIRE: DecoupledWorkerSynthesizer, LanguageToolchainDriver

SD7-2050-NSV-02+SMT-equivalence-proofs [2050-NEURO-SYMBOLIC]

Track: B

dependsOn: [SD7-2050-NSV-01]

R: Research Z3 / SMT-LIB equivalence proofs on bytecode or AST that discharge arithmetic overflow freedom, array bounds, null dereference impossibility, state-machine validity.

I: $\forall x \in \text{Domain}(\text{requires}): \text{Implementation}(x) \Rightarrow \text{ensures}(x)$. Failure to discharge triggers repair loop before test suite. Foreseeable error: solver timeouts; set strict resource bounds and fail closed.

WIRE: GovernedCompileGate, CompilerDiagnosticRepairLoop**SD7-2050-NSV-03+tri-model-disputation** [2050-NEURO-SYMBOLIC]**Track:** S**dependsOn:** [SD7-T1-040]**R:** Research heterogeneous multi-model synthesis with structural AST edit-distance comparison and adversarial arbitration for high-risk logic.**I:** High-risk state-mutating algorithms synthesized by three independent families. If AST semantics diverge, arbitrator extracts discrepancy and uses SMT oracle to select the superior branch. Foreseeable error: majority vote without proof; always require SMT discharge.**WIRE:** ProviderCascadeRouter, DecoupledWorkerSynthesizer, VerifiedCompletionGate**4.4 Long-Horizon Residual Defect Elimination****SD7-2050-LHD-01+concolic-test-generator** [2050-NEURO-SYMBOLIC]**Track:** B**dependsOn:** [SD7-T1-030, SD7-2050-NSV-02]**R:** Research concolic (concrete + symbolic) test generation that forces exploration of unreached branches via SMT path-condition inversion.**I:** Target 100 % branch and condition coverage (MC/DC for critical lanes). Foreseeable error: path explosion; bound depth and record uncovered residual as Residual Obligation.**WIRE:** Wave1 test generator, VerifiedCompletionGate**SD7-2050-LHD-02+transactional-memory-leak-detector** [2050-NEURO-SYMBOLIC]**Track:** B**dependsOn:** [SD7-2050-FMC-02]**R:** Research byte-accurate heap and descriptor profiling that proves allocations – deallocations = 0 per transaction after warmup.**I:** Any non-zero growth of unreferenced bytes rejects the build as resource-unsound. Foreseeable error: false positives from caches; exclude known warm caches from the calculation.**WIRE:** Digital Twin runner, EvidenceCollector**SD7-2050-LHD-03+residual-obligation-ledger** [2050-NEURO-SYMBOLIC]**Track:** S**dependsOn:** []**R:** Research immutable CAS-chained ledgers of human vetoes that future synthesis engines treat as hard boundaries.**I:** Every human rejection creates a durable formal prohibition clause. Future DAG synthesis reads Residual Obligations as hard constraints. Foreseeable error: loss of vetoes across restarts; store only as CAS-addressed records with exact hashes.**WIRE:** LocalMemoryStore, CAS, RestartCoordinator**SD7-2050-LHD-04+zero-entropy-proof-certificate** [2050-NEURO-SYMBOLIC]**Track:** S**dependsOn:** [SD7-T1-060, SD7-2050-NSV-02, SD7-2050-FMC-01, SD7-2050-FMC-02]**R:** Research machine-checkable cryptographic proof bundles that third parties can validate in seconds without trusting the LLM.**I:** Final deliverable is a certificate vector: SpecHash, ExpandedNegativeSpecHash, SMTVerificationProofsHash, TLAModelCheckReceiptHash, MutationKillScoreHash, DigitalTwinChaosReceiptHash, ToolchainBinaryHash. Foreseeable error: missing fields; VerifiedCompletionGate refuses promotion until all hashes are present.**WIRE:** VerifiedCompletionGate, EvidenceCollector, IntegrityProofs**5. File Constitution & Anti-Boilerplate Atoms (Tier-1)****SD7-T1-070+file-atomicity-constitution** [NEW-SUBSTRATE]**Track:** S**dependsOn:** []**R:** Research single-responsibility file limits and composition-over-inheritance enforcement used in large codebases that survive multi-year autonomous evolution.**I:** Soft 350 LOC / hard 500 LOC. One file = one atomic responsibility. No mixing presentation with decision or execution logic. Enforced by ArchitectureComplianceChecker as deterministic gate. Ban TODO, NotImplemented, fake returns. Foreseeable error: generated files that exceed cap; reject before merge.**WIRE:** ArchitectureComplianceChecker, DecoupledWorkerSynthesizer**SD7-T1-071+AST-stub-hollow-scanner** [NEW-SUBSTRATE]**Track:** S**dependsOn:** [SD7-T1-070]**R:** Research static AST walks (Tree-sitter or language-specific) that reject hollow implementations with zero false negatives on the listed patterns.**I:** Hard rejection of TODO(), NotImplementedError, empty function bodies, empty catch blocks, constant returns, unused parameters in business logic, throw-fake-exception, MockClass in production path. Synthesis rejected immediately, zero reward. Foreseeable error: language-specific AST gaps; start with Kotlin + TypeScript + Python + Rust.**WIRE:** ArchitectureComplianceChecker, DecoupledWorkerSynthesizer**SD7-T1-072+mutation-testing-oracle** [NEW-SUBSTRATE]**Track:** B

dependsOn: [SD7-T1-030]**R:** Research systemic defect injection (negate conditionals, swap operators, invert boundaries, erase calls) and the 85 % kill-rate threshold.**I:** Before atom acceptance inject ≥ 10 mutations. Wave-1 suite must fail on ≥ 85 %. Passing mutated code proves the test is fake and rejects both module and test. Foreseeable error: equivalent mutants; record and quarantine them.**WIRE:** Wave1 test generator, VerifiedCompletionGate

6. Mandatory Caller-Completeness Gate

After all Tier-1 atoms are implemented, execute the following audit before any promotion:

1. Every new class introduced by this DAG (AutonomousWaveOrchestrator, PartitionClusterManager, ContractLatticeSynthesizer, DecoupledWorkerSynthesizer, DependencyInjectionAssembler, NativeToolchainRegistry, all LanguageToolchainDriver implementations, CompilerDiagnosticRepairLoop, and every 2050 atom owner) must appear in at least one production WIRE outside tests.
2. AppSourceTemplate.kt and LanguageScaffold.kt must have zero remaining references in production source.
3. ArchitectureComplianceChecker must report zero violations of the file-atomicity constitution and the AST-stub scanner.
4. A real application (CLI or web) must be synthesized end-to-end through AutonomousWaveOrchestrator and pass VerifiedCompletionGate with a non-empty artifact.
5. The Zero-Entropy Proof Certificate must be present and all hashes non-null.

Failure of any item blocks promotion. This gate is itself an atom:

SD7-T1-099+caller-completeness-gate [NEW-SUBSTRATE]**Track:** S**dependsOn:** [ALL Tier-1 atoms]**R:** Research orphan-detection and caller-graph completeness checks used in large monorepos (Google, Meta, Microsoft) that prevent dead code and unwired new modules.**I:** Implement as an atropos doctor --factory-completeness command that fails CI if any of the five conditions above is false. Foreseeable error: test-only callers counted as production; require callers outside src/test.**WIRE:** CI job, atropos doctor, ArchitectureComplianceChecker

7. Critical Path (Tier-1)

SD7-T1-001 $\rightarrow$ 002 $\rightarrow$ 003 $\rightarrow$ 004SD7-T1-010 $\rightarrow$ 011 $\rightarrow$ (012|013|014|015|016|017) [drivers may proceed in parallel]

SD7-T1-018 (PartitionClusterManager)

SD7-T1-020 $\rightarrow$ 021 $\rightarrow$ 030 $\rightarrow$ 040 $\rightarrow$ 041 $\rightarrow$ 050 $\rightarrow$ 060

SD7-T1-061, 062, 063, 064, 065 may proceed once their direct dependsOn are green.

2050 atoms that depend only on Wave-0 may run in parallel with Wave-2/3 once contracts exist.

SD7-T1-099 is the final gate.

8. Non-Negotiable Mathematical Invariants

 $I(\text{Module_A}; \text{Module_B} \mid \text{Interface_Lattice}) = 0.000 \text{ bits}$ $\text{Territory}(\text{P_r}) \cap \text{Territory}(\text{P_s}) = \emptyset$

Non-zero native exit code = terminal rejection (zero false positives)

 $\text{Worker context} \leq \text{partition footprint}$ (default $\leq 12\,000$ tokens) $C_{\text{real}} = \min(C_{\text{impl}}, C_{\text{int}}, C_{\text{ver}}, C_{\text{ev}})$

False-Green Tolerance = 0.0000 %

Every code byte originates from atomic LLM synthesis against formal interfaces verified by genuine native compilers.

9. Acceptance Criteria for Tier-1 Completion

- AppSourceTemplate.kt and LanguageScaffold.kt are deleted and have zero production references.
- AutonomousWaveOrchestrator can synthesize a non-trivial multi-file application from SpecGraph atoms.
- All generated files pass ArchitectureComplianceChecker (atomicity + stub scanner).
- Real toolchain drivers are invoked; no mock verify.sh remains.
- At least one 2050 neuro-symbolic gate (ILSS or FMC or NSV) is live and blocking.
- Caller-Completeness Gate (SD7-T1-099) is green.
- A Zero-Entropy Proof Certificate is emitted for the synthesized application.

Only after the above is true may Tier-2 atoms be scheduled.

